DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

B.Tech (AI & ML) Academic Assignment

WEB DEVELOPMENT PROJECT REPORT

A Comprehensive Study on Front-End Web Design using HTML5 and CSS3

STUDENT SUBMISSION DETAILS

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Subject:	Web Development
Technologies:	HTML5 & CSS3
Academic Year:	First Year B.Tech (2026)

Submitted for Academic Assessment in Web Development Course

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Table of Contents

1. Introduction to Web Development	Page 2
2. Technologies Used — HTML5 & CSS3	Page 2
3. Project 1: Personal Portfolio Website	Page 3
4. Project 2: Fitness Center and Gym Website	Page 4
5. Project 3: Moon Light Restaurant and Cafe	Page 5
6. Conclusion & Key Learning Outcomes	Page 6

1. Introduction to Web Development

Web development is the process of creating and maintaining websites and web applications that run in web browsers. It can involve front-end design, back-end server management, and database configuration. Understanding front-end web design is a fundamental skill for computer science students.

This report describes three practical web development projects created around common website needs. The projects demonstrate page structure, visual layout, responsive design, navigation, and user-friendly interface elements using HTML5 and CSS3.

2. Technologies Used

Technology	Role in Web Development	Key Features Applied
HTML5 (HyperText Markup Language)	Provides the structure and content of a webpage.	Semantic tags such as header, section, footer; headings, lists, links, images, and forms.
CSS3 (Cascading Style Sheets)	Controls visual presentation, layout, colors, and spacing.	Flexbox, Grid, card designs, hover effects, responsive layouts, typography, and transitions.

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Project 1: Personal Portfolio Website

1.1 Introduction

A personal portfolio website helps an engineering student present academic details, educational background, technical skills, hobbies, projects, achievements, and contact information in one place.

1.2 Aim

To design a responsive Personal Portfolio Website using HTML5 for structure and CSS3 for visual styling.

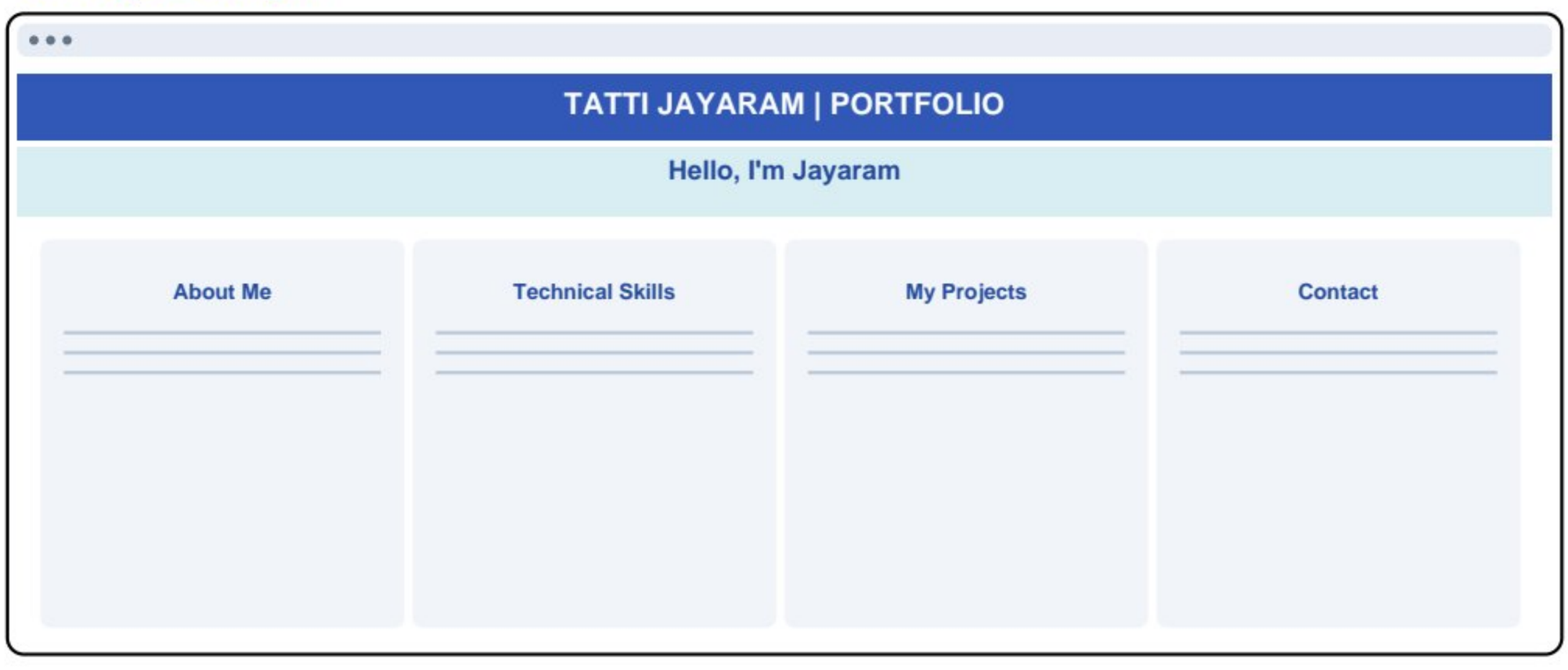
1.3 Key Features

- **Navigation Menu:** Links to Home, About, Skills, Projects, and Contact sections.
- **Profile Banner:** Displays the student's name, program, and a resume link.
- **Educational Qualifications:** Shows degree, branch, and college information.
- **Skills & Hobbies:** Lists programming skills such as Python, HTML, and CSS, along with hobbies.
- **Projects & Achievements:** Presents completed mini-projects and accomplishments.
- **Contact Section:** Provides contact information and links styled with CSS.

1.4 Usage Instructions

Open the file *Personal Portfolio Website.html* in a browser such as Chrome, Edge, or Firefox. Alternatively, open the project in VS Code and use Live Server to preview the webpage.

1.5 Project Output



Output 1: Personal Portfolio Website preview

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Project 2: Fitness Center and Gym Website

2.1 Introduction

The Fitness Center and Gym Website introduces a fitness facility and presents its workout programs, training options, membership plans, facilities, and class schedules. It helps visitors understand the available services before joining.

2.2 Aim

To create a responsive gym website using HTML5 and CSS3, with clear sections, bold typography, program cards, pricing tables, and registration forms.

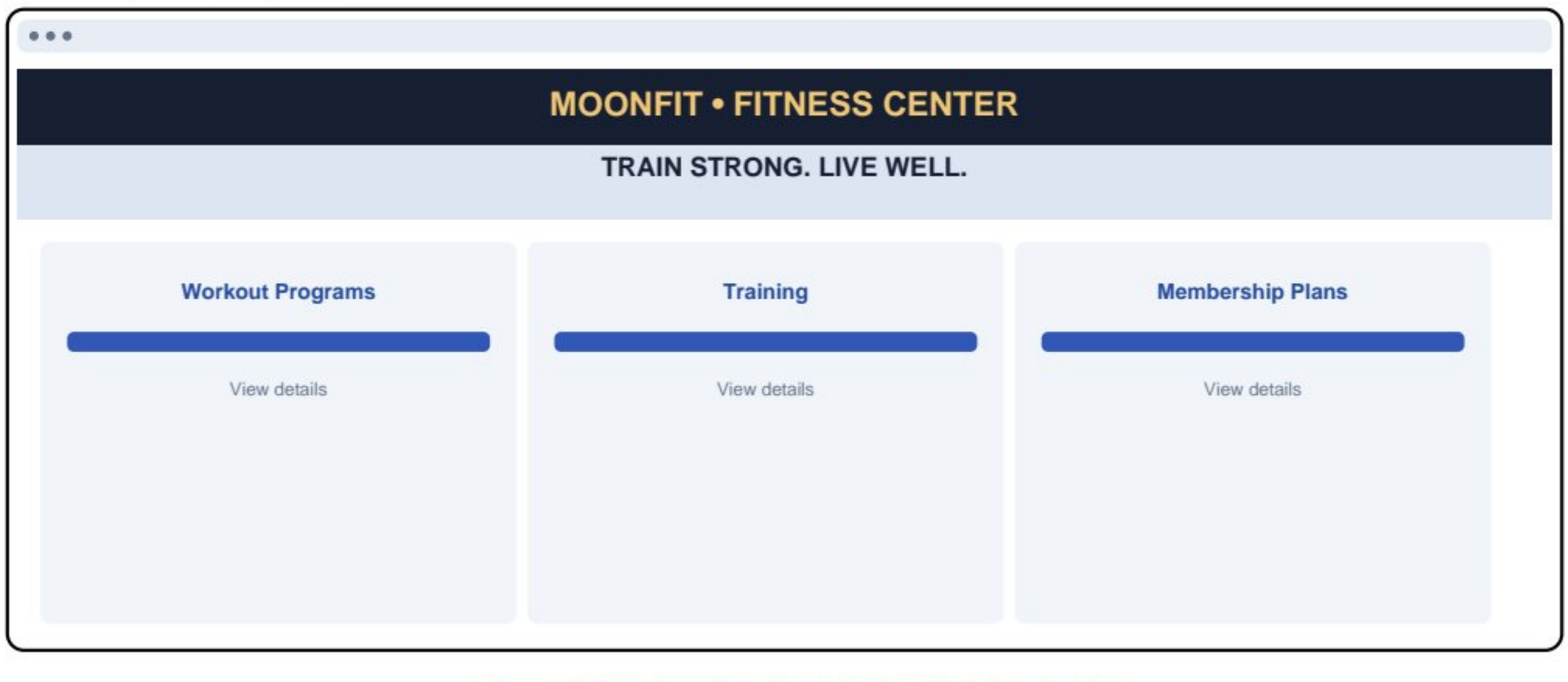
2.3 Key Features

- **Welcome Banner:** A bold introduction with gym branding and navigation links.
- **Workout Programs:** Sections for strength training, cardio, flexibility, and general fitness.
- **Personal Training:** Information about trainer support and guided workout sessions.
- **Membership Plans:** Responsive pricing cards showing plan names, duration, and fees.
- **Fitness Facilities:** Details of workout equipment, training areas, and available amenities.
- **Class Schedule:** A simple weekly timetable for group sessions.
- **Registration Form:** HTML form fields for name, contact details, and preferred membership.
- **CSS Effects:** Responsive layout, hover transitions, and clear visual contrast; no JavaScript is required for the static design.

2.4 Usage Instructions

Open *Fitness Center and Gym Website.html* in a web browser. Visitors can review programs and membership options, check the class timetable, and fill in the sample registration form. The form is a front-end demonstration; storing registrations requires a connected backend.

2.5 Project Output



Output 2: Fitness Center and Gym Website preview

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Project 3: Moon Light Restaurant and Cafe

3.1 Introduction

The Moon Light Restaurant and Cafe website presents a welcoming online identity for a restaurant and cafe. It highlights menu categories, sample prices, chef information, customer reviews, opening hours, location details, and table reservation requests.

3.2 Aim

To build a themed, multi-section restaurant website using HTML5 and CSS3, with a warm color palette, responsive menu cards, and a simple reservation form.

3.3 Key Features

- **Brand Header & Welcome Section:** Moon Light branding, introductory text, and navigation links.
- **Categorized Food Menu:** Cards for Starters, Main Course, Desserts, and Drinks, with prices in Indian rupees.
- **Chef Profiles:** Short introductions to the kitchen team and featured dishes.
- **Customer Reviews:** Testimonial cards to demonstrate customer feedback.
- **Table Reservation Form:** Date, time, guest count, name, and contact fields.
- **Location & Hours:** Weekly opening hours and restaurant contact details.
- **Responsive Styling:** CSS layout adjustments, readable typography, and hover effects.

3.4 Usage Instructions

Open *Moon Light Restaurant and Cafe.html* in a browser. Visitors can browse the menu, read the sample reviews, check opening hours, and submit a sample table-reservation request. Processing real bookings would require backend support.

3.5 Project Output



Output 3: Moon Light Restaurant and Cafe preview

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6. Conclusion & Key Learning Outcomes

Through these three front-end web development projects, I practiced building webpages with HTML5 and CSS3. The portfolio, gym website, and restaurant-and-cafe website demonstrate how structure, styling, and layout can be adapted to different purposes and audiences.

Summary of Skills Acquired

- **Semantic HTML5 Structure:** Used meaningful page sections such as header, section, article, and footer.
- **CSS Layout Techniques:** Applied layout concepts including Flexbox, Grid, spacing, and alignment.
- **Responsive Design:** Organized content so that it can adapt to different screen sizes.
- **Form Design:** Practiced creating labeled input fields and basic browser validation.
- **UI/UX Enhancement:** Used visual hierarchy, button styles, hover transitions, and suitable color palettes.
- **Project Workflow:** Practiced organizing HTML/CSS files and previewing pages in a browser or VS Code Live Server.

Student Declaration

I hereby declare that the web development projects presented in this report were prepared as part of my first-year B.Tech coursework. The report describes front-end website designs using HTML5 and CSS3.

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